Loan Portfolio Analytics Report

1.Problem Statement

The objective of this analysis is to evaluate loan performance, identify default trends, and understand borrower behavior using the Lending Club dataset (2007–2016). The goal is to uncover risk patterns and support better lending decisions.

2.Dataset Overview

- **Source: Lending Club Dataset**
- **Time Period: 2007–2016**
- **Key Fields:**
  - Loan Status (Fully Paid, Charged Off)
  - Loan Amount
  - Interest Rate
  - Term (36 / 60 months)
  - Grade (A–G)
  - Home Ownership
  - Loan Purpose

3.Key KPIs

- Total Loan Portfolio: $4.94 Billion
- Average Loan Amount: $13.19K
- Average Interest Rate: 13.55%
- Default Rate: 19.58%

Nearly 1 out of 5 loans defaulted — this is a significant risk indicator.

4.Loan Performance Analysis

- Default vs Fully Paid

1. Fully Paid: 80.42%
2. Default (Charged Off): 19.58%

Portfolio is stable overall, but default rate is still high enough to require risk control.

5. Borrower Profile Insights

- **Home Ownership**
  1. Mortgage: ~49% (highest)
  2. Rent: ~41%
  3. Others: very small share

- **Most borrowers either:**
  1. Already own homes (mortgage)
  2. Or are renters → potential higher risk group

- **Loan Term Distribution**
  1. 36 months: ~77%
  2. 60 months: ~23%

6. Loan Trend Over Time

- Strong growth from 2011 to 2014
- Peak around 2014
- Decline after 2015

7. Loan Purpose Analysis

- Debt Consolidation (highest)
- Credit Card payoff
- Home Improvement

8. Analysis Insight

- **High default rate (~20%)**
- **Debt consolidation dominance**
- **Renters segment significant**
- **Longer-term loans (60 months) → potential higher default risk**

9. Business Recommendations

1. **Stricter Credit Screening**
   - Especially for debt consolidation loans
2. **Interest Rate Adjustment**
   - Renters
   - Long-term loan applicants
3. **Interest Rate Adjustment**
   - Risk-based pricing for lower-grade borrowers
4. **Limit Long-Term Loans**
   - Reduce exposure to 60-month loans

10. Conclusion

The loan portfolio shows strong growth and stable repayment behavior, but a ~20% default rate highlights significant credit risk. By focusing on borrower segmentation, loan purpose, and term structure, financial institutions can improve portfolio quality and reduce losses.